

Core: Requirements

CHAPTER 4: REQUIREMENTS MANAGEMENT

Chapter Objectives

In this chapter, we will understand requirements change management by:

- Discussing the reasons requirements change
- Identifying the components of a change management process
- Clarifying the importance of requirements traceability
- Examining traceability in an Agile SDLC

Chapter Concepts

The Inevitability of Change

Traceability

Agile Change Management

Activity 4.1: The Moving Goalpost Challenge

Experiencing Change Under Pressure



Goal	Experience how repeated change creates pressure, rework and resistance — and what helps a team absorb it
Team Format	Teams of 4–6
The brief	You are the requirements team for Buy Local Bonds. Release 1 must be baselined in four weeks. Produce a one-page baseline showing: – Six well-formed, testable requirements for Release 1 – The type and category of each — BR/UR/SR and A/B/C/D/FR/NF – A MoSCoW priority for each – How you have organised them — by usage, audience or quality – The document you will baseline them in Three minutes to start. Then the goalposts move.
Deliverable	A revised Release 1 baseline and a 60-second readout
Duration	20–25 minutes

Activity 4.1: The Goalposts Move



- 1. Two weeks, not four.** Leadership has pulled the launch date in.
- 2. The analyst team is halved.** Whatever you document, you document with fewer people.
- 3. Legal has ruled.** Customers must acknowledge a suitability disclosure before any bond price is displayed.
- 4. The brokerage heads object.** Direct purchase cuts their commission. They want the “without a broker” requirement removed.
- 5. The measure has changed.** Leadership has stopped counting transactions. They now want a documented reduction in call-centre volume within 30 days.

Revise your baseline after every announcement.

Activity 4.1: Debrief



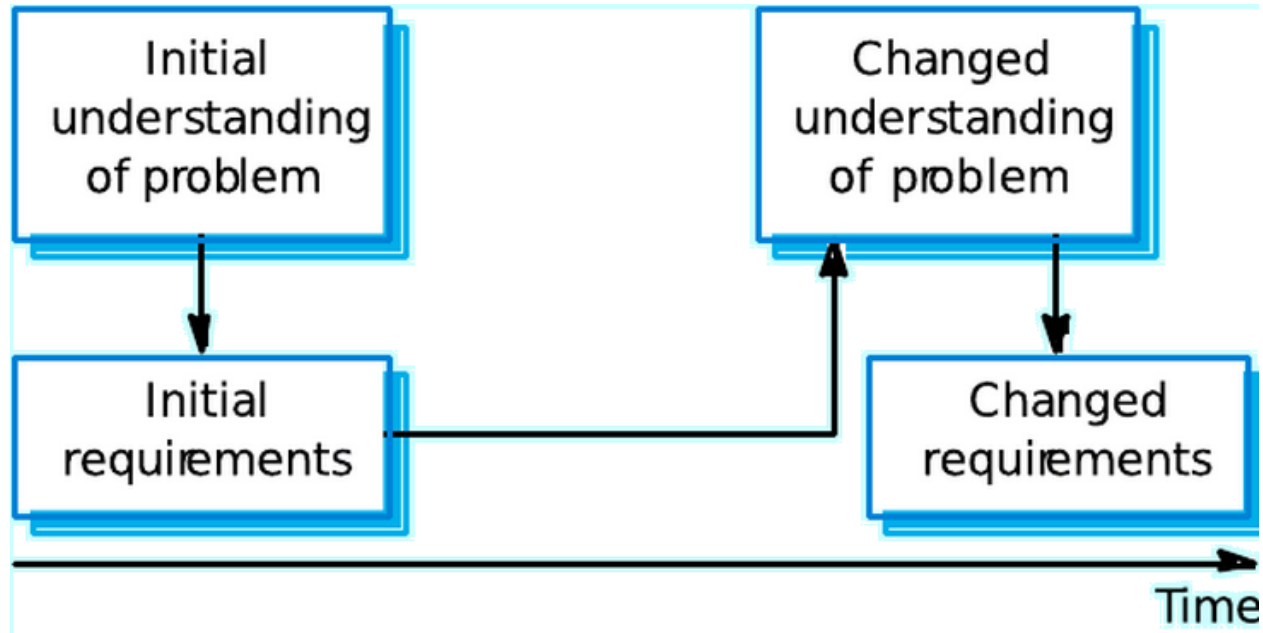
Sixty seconds per team: your final baseline · what you abandoned or redesigned · which change created the most pressure · what helped, or would have helped

Then discuss

- What was your first reaction to each announcement?
- When did the team become frustrated or resistant?
- Did you keep patching the original baseline, or step back and redesign it?
- What happened to communication and decision-making under pressure?
- Which changes needed clarification rather than immediate action?
- What would a leader have needed to give you to make this manageable?

Key learning People rarely resist change itself. They resist uncertainty, loss of control, repeated rework, and change that arrives without explanation.

The Cycle of Change



Requirements Evolve

- ◆ Requirements should be specified as completely as possible at the point in time they are created
- ◆ Change is inevitable
 - Evolve the requirements as more is learned
- ◆ A formal change process should be used to help manage changing requirements
 - Identify, control, track, and report projected changes
- ◆ Approved changes in requirements lead to changes in the requirements documentation
 - Provide an accurate audit trail of what has changed over time
 - A prior version of a requirement should be available for review at any point in a project



Requirements Management



- ◆ Requirements management is a process that establishes and maintains agreement between the customer and the project team on the changing requirements of the system
- ◆ Manages change throughout the life cycle
 - Expect an average 25 percent change or growth in unanticipated requirements in a typical two-year project
- ◆ Change management is typically a project manager's job
 - Analysts interact with customers during development to drive change
 - Testers interact with test results that often drive change

Requirements Configuration Management

- Configuration management (CM) of requirements is important for the same reasons that configuration management of source code is important
 - Software Configuration Management (SCM) pays particular attention to managing the QA and Production coding environments
- Benefits of requirements management
 - Prevent unauthorized change
 - Preserve revisions of requirements documents
 - Retrieve previous versions of documents
 - Allow a managed baseline release strategy
 - Prevent simultaneous update of documents
- Helps coordinate changes in a collaborative environment of multiple designers and developers working on the same artifacts



A Good Change Management Process



- A good change management process does five things
 1. Communicates the intent of the change
 2. Manages the processing of the change request
 3. Enables the analysis of the impact of the change
 4. Tracks the implementation of the change
 5. Communicates implemented changes to all stakeholders

Change Requests

- ◆ Change requests have two major sources
 - Defects
 - Test feedback
- ◆ Change requests can be created during the development process and after a product is in production
- ◆ Accurate impact statements by the project team are essential to the successful management of change
 - What is the change going to cost in terms of labor hours?
 - What is the impact on the rest of the iteration's deliverables?
 - What does it do to the priorities?
 - Will anything have to be given up?



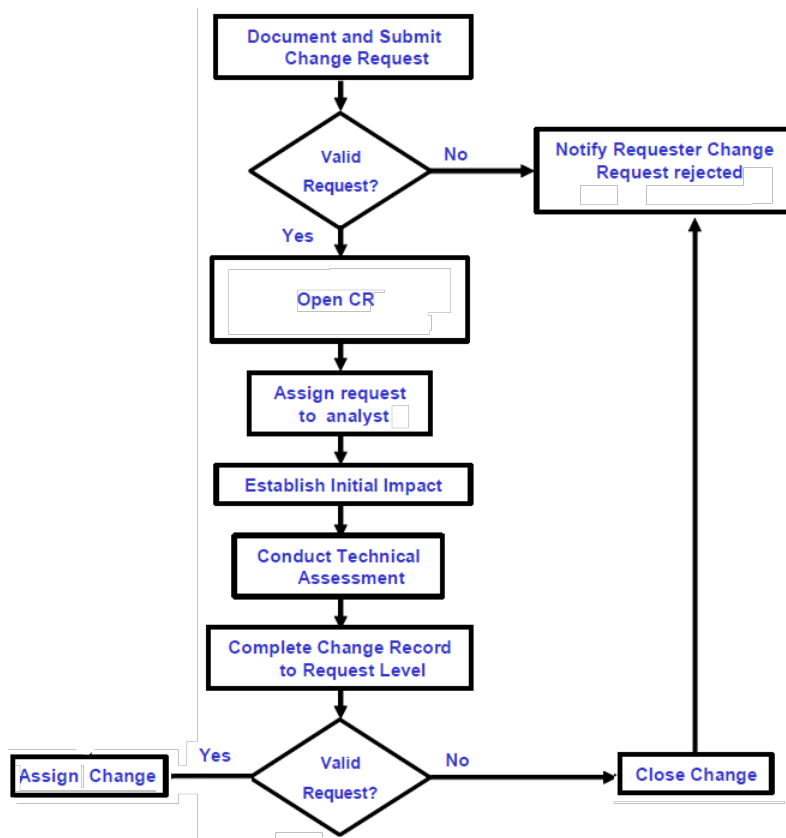
In Scope or Out of Scope?



Scoping

- The analyst's first job relative to any proposed change is to ensure it is in scope
- Compare the change to the scope statement
- Compare the change to the company's mission
- Compare the change to the company's strategic direction
- For out of scope changes, master the statement
 - "This is a nice idea, but it's outside the scope of the current effort. Do you want to put in a formal change request to change the project scope?"

An Example of a Formal Change Management Process

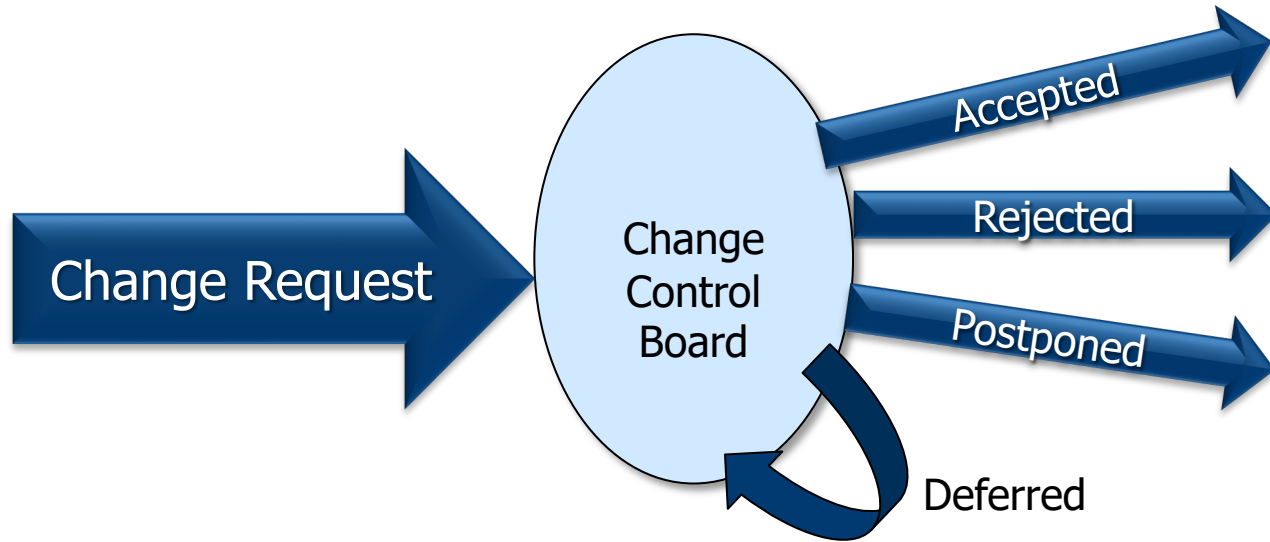


Impact of Change

- ◆ Before a decision can be made, impact of change must be evaluated
 - How long?
 - At what cost?
 - Impact on other activities
 - Risk associated with change
 - Impact on overall project
- ◆ The SA, QA along with the Project Manager are responsible for this impact assessment
 - Use Impact Risk Analysis
 - Use 3-point estimating
 - Use brainstorms if required
- ◆ A decision cannot be made if the impact cannot be evaluated



Disposition of Change



Activity 4.2: In Scope or Out of Scope



Goal	Decide if a change is in or out of scope
Team Format	Case study teams
Materials	This slide
Tasks	1. The driving force behind BLB has decided that the system needs to include features that allow a customer to understand a management risk in their bond portfolio 2. Consider whether this is in scope or out of scope 3. Be prepared to defend and communicate your choice
Deliverable	A well-considered decision on in-scope or out of scope
Duration	20 minutes

Chapter Concepts

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Traceability

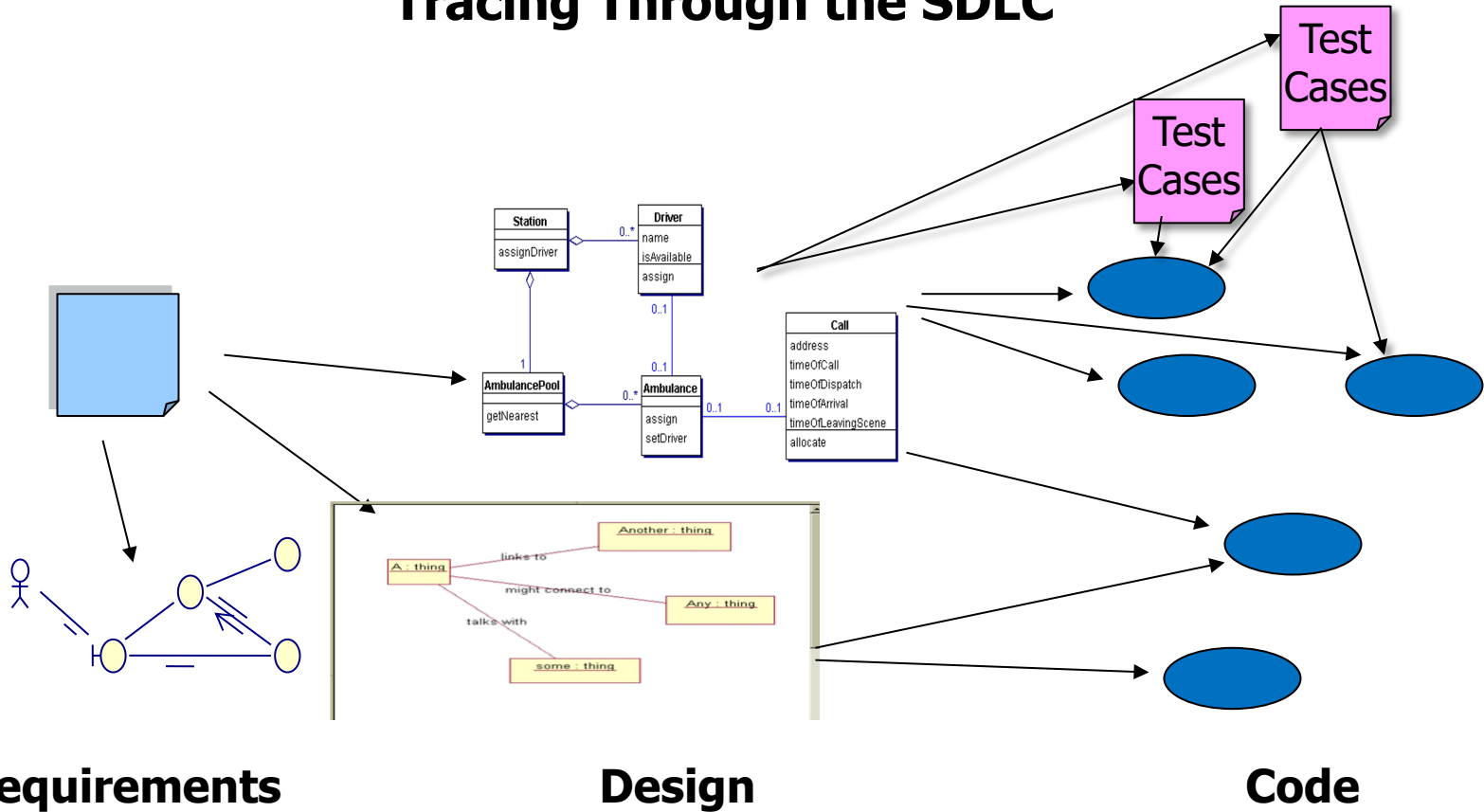
Agile Change Management

Traceability

- ◆ The chance of change occurring during development is just about 100%
- ◆ Change management ensures...
 - Management will review all changes
 - ◆ Valid changes will be permitted
 - ◆ Unapproved changes will be prohibited
 - All changes meet user requirements
 - Changes
 - ◆ Are only made with appropriate authorization
 - ◆ Occur at the right time
 - ◆ Are tracked from inception to completion
 - ◆ Can be traced to documentation (requirements, test cases)
- ◆ Traceability helps to manage controlled change
- ◆ Discussion topic: Is traceability required in Agile? Discuss pros, cons, and implementation



Tracing Through the SDLC



Tracing the Requirements (continued)



Access Policy

Policy Information will be available

12.4 The policy information will be accessed
12.4.1 by entering policy number
12.4.2 by entering customer id

133.1.1 Policy Access Screen

133.1.1.1 Policy number must be 10 numeric characters

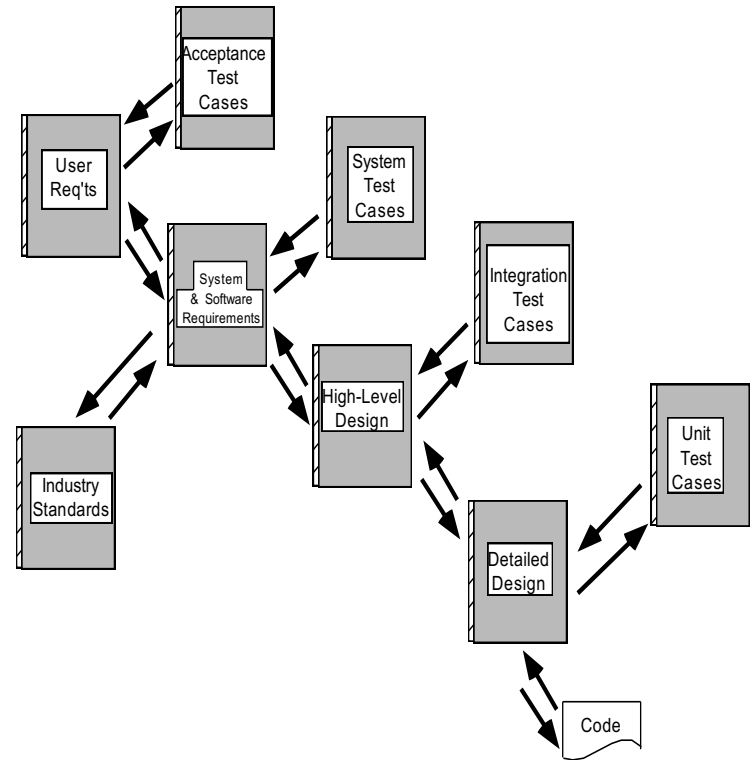
133.1.1.2 When no match, display "Policy cannot be found"



Screen shots/diagrams

The Traceability Matrix

- Confirmation to check that each phase follows the other appropriately
 - Links one phase to another
- Impact analysis for developers to determine full cost of making a change
 - Perform Technical Impact Analysis
- Quality Control to link test cases to sources
 - Can ensure all items are addressed
- Management tool to prove completeness and prevent “scope creep”
 - Each requirement has been implemented in design and code
 - Design and code are related to problem



Traceability Matrix Example

Project Requirements		Item Requested?	Design				Build/Development		
			In Delivered Solution?	In Custom Interface?	In Customization per GAP analysis?	Accounted for in Design?	RFC Req?	Development Complete?	Unit Tested?
A.	Requirement 1 - CHECK-IN FUNCTIONALITY - I.D.A. must provide role-based users with "Quick Check-In" functionality (Closing an order) for iO provisioning overflow calls.	Deferred to 2014	Yes	Yes	N/A	Yes			
B.	Requirement 2 - CLEARTRAN ACCOUNT LOCK ISSUES - When ClearTran is launched as a payment option in the payment screen, I.D.A. must close the payment screen, force the user to close out additional open interactive screens and bring the user back to the 'Interaction Home' page prior to launching ClearTran.	Development	Yes	Yes	N/A	Yes			
C.	Requirement 3 - LINK TO ALL OTHER ACCOUNTS AT CORP/HOUSE - I.D.A. must provide the user with a tab that contains links to all other accounts at that corp/house (active and inactive).	Development	Yes	Yes	N/A	Yes			
D.	Requirement 4 - IGNORE CERTAIN WRAP CODES FOR REPEAT CALLER INDICATOR - The repeat caller indicator logic in I.D.A. must be modified to ignore specific configurable Inbound/Outbound Reason 1, 2 or 3 wrap-up codes when determining repeat caller criteria to eliminate false customer contacts.	Development	Yes	Yes	N/A	Yes			
E.	Requirement 5 - EASTERN TIME ZONE - I.D.A. must default to the Eastern Time zone in the sign on screen.	Development	Yes	Yes	N/A	Yes			
F.	Requirement 6 - BOX HIT STANDARDIZATION - The 'Equipment Overview Page' must be revised to present the box hits identically to those presented in the 'Box Authorization' tool in the repair page (with explanation what to use them for).	Development	Yes	Yes	N/A	Yes			
G.	Requirement 7 - BUSINESS RULES FOR TC TASK CODES - A new business rule must be created to automatically add 'Trouble Call Code' 900 (Bz Acct-Priority SLA) as the first code when scheduling a trouble call on all business accounts. The current Business rule that adds the 400 task code when a repeat T/C is set up must be modified to add the 400 task code if the	Development	Yes	Yes	N/A	Yes			
H.	Requirement 8 - EXISTING SVCS. IN PRODUCT NEGOTIATION PAGE - I.D.A. must prepopulate the customer's existing services as a default when loading the 'Product Negotiation' tab	Development	Yes	Yes	N/A	Yes			
	Requirement 9 - LAUNCH OUTAGE BOARD IN CONTEXT - I.D.A. must								

http://www.ieee.li/tmc/traceability_matrix_sample.pdf

Activity 4.3: Traceability Matrix



Goal	Examine the strengths and weaknesses of an RTM
Team Format	Two teams
Materials	Flip charts
Tasks	1. The instructor will assign one team Strengths (pros) and the other team Weaknesses (cons) 2. Brainstorm the concept of the Requirements Traceability Matrix (RTM) from the assigned perspective 3. Share your results with the class
Deliverable	A debate on the pros and cons of RTMs
Duration	20 minutes

Chapter Concepts

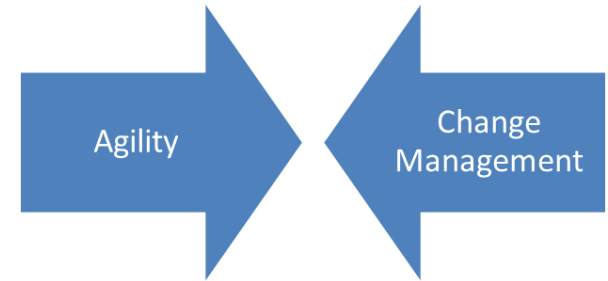
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Agile Change Management

Six Tips for Agile Change Management

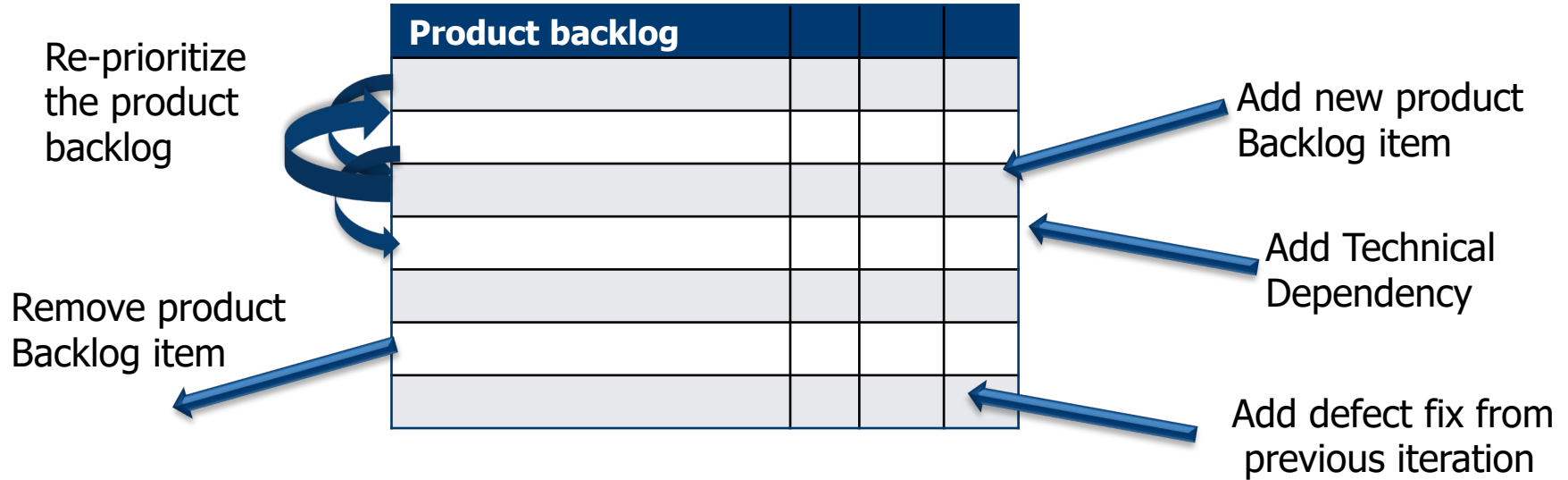
1. The goal of Agile is to enable change management, not prevent it
2. Change management is inevitable in Agile
3. There should be no change management during an iteration
4. Change management does not mean no prioritization
5. Change management does not change highest value story scheduling
6. Agile does not eliminate change management challenges but provides a disciplined, streamlined way to manage them



Agile Change Management

- In Agile there is no separate change management process
 - “Change” is alteration to software
 - May be defect fix, enhancement, or product owner changes mind
- All approved changes are incorporated into product backlog
 - Prioritized by customer into a release
 - Customer makes the trade-off between changing existing functionality or working on new functionality
- Still need all five aspects of a good change management process
 1. Communicates the intent of the change
 2. Manages the processing of the change request
 3. Enables the analysis of the impact of the change
 4. Tracks the implementation of the change
 5. Communicates implemented changes to all stakeholders

Positioning a Change in the Planned Work



Task Board

- ◆ A task board displaying user stories creates a visual illustration of progress and what would be impacted by the placement of a change

Story	To Do		In Process	To Verify	Done
As a user, I... 8 points	Code the... 9	Test the... 8	Code the... DC 4	Test the... SC 6	Code the... D
	Code the... 2	Code the... 8	Test the... SC 8		Test the... SC 8
	Test the... 8	Test the... 4			Test the... SC
As a user, I... 5 points	Code the... 8	Test the... 8	Code the... DC 8		Test the... SC
	Code the... 4	Code the... 6			Test the... SC 6

<http://www.mountangoatsoftware.com>

Chapter Summary

In this chapter, we have developed an understanding of requirements change management by:

- ◆ Discussing the reasons requirements change
- ◆ Identifying the components of a change management process
- ◆ Clarifying the importance of requirements traceability
- ◆ Examining traceability in an Agile SDLC